Department of Computer Science

USABLE SECURITY AND PRIVACY
COMP438

Project Proposal

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1. Core Idea (Summery):

Our project is a shoulder-surfing detector designed for blind and visually impaired users who rely on **TalkBack** on Android. When TalkBack is about to read sensitive information like passwords or PINs, it sends a signal to our application. The app then activates the camera, checks if someone is nearby, and tells TalkBack whether to pause or continue speaking to protect the user's privacy.

2. Project Title:

SecureRead TalkBack Helper

3. Problem Statement:

Blind and visually impaired users depend on mobile screen readers like TalkBack to read passwords, PINs, messages, and other private information out loud. When they are in public or shared spaces, people around them may overhear this sensitive content or see it by looking at the screen from behind. This creates a serious security and privacy risk known as **shoulder surfing**.

Shoulder surfing is common because everyday tasks require entering passwords or listening to personal information through TalkBack. Visually impaired users cannot easily detect if someone is standing close to them or observing their device. They need a simple, automatic way to know if it is safe for TalkBack to speak sensitive information.

4. Main Features:

- **Signal From TalkBack Add On**

The TalkBack accessibility service detects predetermined sensitive keys or words (such as passwords, PINs, or private information) and sends a message to our application.

- **Early Warning to the User**

When the message is received, the application warns the user that TalkBack is about to read sensitive information. The user can choose to continue normally or to check the surroundings first.

- **Surrounding Check Mode**

If the user chooses to check the area, the app activates the phone's camera and uses AI (image processing, computer vision, and human detection models) to scan the environment.

- **Shoulder-Surfing Detection**

The system checks if there are people near the user who may overhear the sensitive information. If someone is detected, the app alerts the user about possible shoulder-surfing risks.

- **Environment Feedback**

The application describes what it sees in simple language, such as whether the area is clear or if one or more people are around the user.

- **User Control Over TalkBack**

After hearing the feedback, the user can choose to stop TalkBack from reading the sensitive content or continue safely if the area is clear.

5. Related Applications:

There are several applications and research tools that focus on shoulder-surfing protection or environment awareness, but none of them work with TalkBack or react to sensitive spoken content on Android. The closest examples are:

1. Yarosh et al. Shoulder Surfing Detection System (Research Prototype)

A computer vision system that uses a device camera to detect when another person is standing behind the user.

- It alerts the user about possible shoulder surfing.
- However, it is built for sighted users and is not integrated with mobile screen readers like TalkBack.

2. Android Privacy Guard & "Second Face Detection"

Some Android devices warn the user when a second face appears in the camera view.

- This protects visual privacy in apps and notifications.
- But it does not help blind users, and it does not connect to TalkBack or spoken sensitive information.

3. Computer Vision Human-Detection Models (YOLO, OpenCV)

These AI models detect humans in camera frames and are used for surveillance, monitoring, and environment-awareness features.

- They provide reliable detection of nearby people.
- But they are not linked to TalkBack and do not respond when sensitive information is about to be spoken aloud.

4. Privacy Blur & Shoulder Surf Stopper Apps

Apps like PrivacyBlur, ScreenShield, and Shoulder Surf Stopper blur the screen or hide content when someone is detected.

- They help protect visual content.
- They are not suitable for blind or visually impaired users who rely on audio instead of screen visibility.

5. Seeing AI (Microsoft)

A well-known accessibility app that uses AI to describe scenes, objects, and people to blind users.

- It provides environmental awareness.
- But it does not automatically activate when sensitive information is being read by TalkBack.

6. TalkBack "Password Speak Mode"
TalkBack allows users to choose whether passwords should be spoken or only spelled character by character.

- This improves privacy in some cases.
- But it gives no information about the user's surroundings and does not detect shoulder surfing risks.

These tools provide pieces of the solution, visual privacy, human detection, or environment descriptions, but none combine AI-based detection with TalkBack's audio output or offer real time protection when sensitive information is about to be spoken. This makes SecureRead TalkBack Helper a unique and accessible solution.

6. Added Value:

SecureRead TalkBack Helper provides protection that current Android privacy features and accessibility tools do not offer. While many smartphone security systems focus on visual privacy, such as blurring the screen or detecting extra faces, **they do not protect audio output**, which is the main way blind and visually impaired users receive information through TalkBack. Our solution activates **exactly when TalkBack is about to speak sensitive information**, such as passwords, PINs, or private messages. This timing based reaction makes it more accurate and more helpful than general environment aware apps. The system also uses the phone's camera to check the surroundings and warn the user if someone is nearby, helping to prevent shoulder surfing in public places.
After the detection, the app gives **simple audio feedback** about whether the area is safe. The user can then choose to either continue or stop TalkBack from reading the sensitive content. This keeps the experience usable, fast, and fully accessible.
Key Added Values

- **Designed specifically for TalkBack users on Android.**
- **Responds automatically when sensitive content is detected.**
- **Uses camera-based human detection to identify shoulder surfing risks.**
- **Provides clear spoken feedback about the surroundings.**
- **Allows the user to choose whether TalkBack continues or stops.**
- **Addresses audio privacy**, an issue not covered by existing Android features.

7. Platform Selection:

Platform: Android Mobile Application

Our project will be developed as an **Android mobile application** that works alongside **TalkBack**, the main screen reader used by blind and visually impaired users on Android devices. This platform is suitable because it gives access to the phone's camera, system permissions, notifications, and accessibility services, which are needed for detecting sensitive content and checking the surroundings.

Users

The main users are **blind and visually impaired individuals** who rely on TalkBack to read sensitive information such as passwords, PIN codes, confirmation messages, and private data. These users often perform tasks in public or shared environments where people around them can overhear information read aloud.

Technology

The project will use:

- **Android Accessibility Service** to detect when TalkBack is about to read predetermined sensitive words or fields.
- **Android camera APIs** (front/back camera) for capturing frames of the environment.
- **AI models for human detection**, such as TensorFlow Lite or ML Kit, to identify if someone is nearby.
- **Audio feedback** to warn the user about the environment in simple, clear language.
- **User interaction options** (Continue / Stop) through accessible dialogs or spoken prompts.

Context of Use

The application is designed for situations where blind and visually impaired users:

- Are typing passwords or listening to private messages
- Are using their phone in public transportation, classrooms, cafés, workplaces, or open spaces
- Cannot easily know if someone is close enough to overhear sensitive content

In these contexts, our tool adds an extra layer of **audio privacy protection**, helping the user feel more secure and in control.